Notes: Lagos (RES, 2006)

A Model of TFP

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1 Big picture

- The paper builds a *search-and-matching* model where matches draw idiosyncratic productivity x and can be destroyed endogenously.
- Aggregate *TFP is endogenous.* It moves with labor-market institutions because policies shift the *selection margin* (the reservation productivity R) and hence the average productivity among surviving matches.
- With Pareto productivity, the model aggregates to an *exact Cobb–Douglas* production function,

$$Y = AK_e^{\gamma}N^{1-\gamma}, \qquad \gamma = \frac{1}{\alpha}, \qquad A = \frac{R}{1-\gamma},$$

  so TFP A is proportional to the destruction threshold R.
- Measurement matters: if one uses total capital K instead of employed capital K_e, or employment E instead of hours N, the measured Solow residual can move *opposite* to true A because utilization and hoarding respond endogenously.

2 Environment and primitives

2.1 Agents, time, discounting

Continuous time; discount rate r. Mass 1 of workers. A mass of potential firms can post vacancies at cost.

2.2 Matching technology

Let unemployment be u and vacancies be v. Market tightness is $\theta \equiv v/u$. A CRS matching function delivers:

- vacancy filling rate: $q(\theta)$,
- job finding rate: $\theta q(\theta)$.

Separations also occur exogenously at rate δ.

2.3 Match-level production and capital

A firm is a single-job firm. If it is filled, it produces according to the (match-specific) technology (paper Eq. (1))

$$f(x, n, k) = x \min(n, k), \tag{1}$$

where:

- x is match productivity,
- n is hours (chosen within the match),
- k is a firm-level scale/capital choice (chosen when posting the vacancy).

Capital is rented at unit cost c.

2.4 Idiosyncratic productivity shocks

Each match draws a productivity x when formed, and then receives Poisson opportunities to redraw at rate λ. A new draw is i.i.d. from distribution G with support $[\varepsilon, \infty)$.

2.5 A fixed operating cost and "labor hoarding"

There is a fixed cost that makes it possible (sometimes optimal) to keep a match alive but set hours to zero. The paper uses

$$C(x, \varphi) = \max\{\varphi - x,\, 0\}. \tag{2}$$

Intuition: when $x < \varphi$, operating at full scale can be unprofitable, so the match may set $n = 0$ (idle/hoard) while waiting for a higher future draw.

3 Worker and firm problems (value functions)

3.1 Flow profits

Given x, hours n, and scale k, flow profits are

$$\pi(x, n, k) = x \min(n, k) - w - ck - \varphi n - C(x, \varphi)k, \tag{3}$$

where w is the flow wage payment.

3.2 Value functions (baseline, no policy)

Let U be the value of unemployment, $W(x)$ the value of employment in a match of type x. Let V be the value of a vacancy and $J(x)$ the value of a filled job.

The Bellman equations are (paper Eqs. (4)–(7)):

$$rU = b + \theta q(\theta) \int \max\{W(z) - U, 0\}\, \mathrm{d}G(z), \tag{4}$$

$$rW(x) = w(x) + \lambda \int \max\{W(z) - U, 0\}\, \mathrm{d}G(z) - (\delta + \lambda)\big(W(x) - U\big), \tag{5}$$

$$rV = -ck + q(\theta) \int \max\{J(z) - V, 0\}\, \mathrm{d}G(z), \tag{6}$$

$$rJ(x) = \pi(x) + \lambda \int \max\{J(z) - V, 0\}\, \mathrm{d}G(z) - (\delta + \lambda)\big(J(x) - V\big). \tag{7}$$

4 Intra-match decisions: hours and wage bargaining

4.1 Hours choice

Given the fixed proportions in (1), hours optimally satisfy (paper Eq. (8)):

$$n(x) = \begin{cases} k, & x > \varphi, \\ 0, & x \le \varphi. \end{cases} \tag{8}$$

So producing matches operate at full scale $n = k$, while low-x matches may "idle" ($n = 0$).

4.2 Nash bargaining over wages

Wages split the match surplus via Nash bargaining (paper Eq. (9)):

$$(1-\beta)\big(W(x)-U\big) = \beta\big(J(x)-V\big), \qquad \beta \in (0,1). \tag{9}$$

Define total surplus

$$S(x) \equiv \big(W(x)-U\big) + \big(J(x)-V\big).$$

Then (9) implies

$$W(x)-U = \beta S(x), \qquad J(x)-V = (1-\beta)S(x).$$

4.3 Free entry in vacancies

With free entry, firms post vacancies until $V = 0$.

5 Reservation productivity and the structure of surplus

5.1 A key object: the reservation threshold R

A match continues if its surplus is nonnegative. Define R such that

$$S(R) = 0, \qquad S(x) > 0 \text{ for } x > R, \qquad S(x) = 0 \text{ for } x \le R.$$

Thus R is the endogenous job-destruction cutoff.

5.2 Closed-form surplus

Combining the Bellman equations, Nash bargaining, and the hours rule, the paper shows (paper Eq. (14)):

$$S(x) = \frac{x-R}{r+\delta+\lambda}\, k \qquad \text{for } x \ge R. \tag{10}$$

Interpretation: surplus is linear in productivity; the slope is discounted by the effective exit rate $r+\delta+\lambda$.

5.3 Steady-state distribution of match productivities

Let H be the cross-sectional distribution of x among *employed* matches. In steady state (paper Eq. (3)):

$$H(x) = \begin{cases} 0, & x < R, \\ \dfrac{G(x)-G(R)}{1-G(R)}, & x \ge R. \end{cases} \tag{11}$$

So H is a truncated-and-renormalized version of G because low draws are destroyed.

5.4 Unemployment in steady state

From flow balance (paper Eq. (2)):

$$u = \frac{\delta + \lambda G(R)}{\delta + \lambda G(R) + \theta q(\theta)\,[1-G(R)]}. \tag{12}$$

Intuition: inflows are exogenous separations δ plus endogenous destruction events $\lambda G(R)$; outflows are job finding times the probability the match is acceptable.

6 Equilibrium job creation and job destruction

6.1 Job-destruction condition

Imposing $S(R) = 0$ yields a job-destruction (JD) condition (paper Eq. (15)):

$$R - \varphi - c - \left(\frac{b}{k} + \frac{\beta}{1-\beta}c\theta\right) + \frac{\lambda}{r+\delta+\lambda}\int_R^\infty (x - R)\,\mathrm{d}G(x) = 0. \tag{13}$$

6.2 Job-creation condition

Free entry ($V = 0$) gives the job-creation (JC) condition (paper Eq. (16)):

$$\frac{1-\beta}{r+\delta+\lambda}\int_R^\infty (x - R)\,\mathrm{d}G(x) - \frac{c}{q(\theta)} = 0. \tag{14}$$

6.3 Equilibrium definition

An equilibrium is a pair (R, θ) satisfying (13)–(14), plus implied (u, H) via (11)–(12).

7 Aggregation and endogenous TFP

7.1 Employed capital versus total capital

There is capital attached to both filled jobs and vacancies. Let total capital be K. The paper defines employed capital K_e as capital used in filled jobs (paper Eq. (17)):

$$K_e = \frac{1-u}{1-(1-\theta)u}K. \tag{15}$$

7.2 Output and hours

Define the productivity cutoff for *producing* matches:

$$\mu \equiv \max\{R, \varphi\}.$$

Matches with $x \in [R, \varphi]$ survive but set $n = 0$ (labor hoarding). Then (paper Eqs. (18)–(19)):

$$Y = (1-u)\int_\mu^\infty xk\,\mathrm{d}H(x) = [1 - H(\mu)]\,K_e \cdot \mathbb{E}[x \mid x \geq \mu], \tag{16}$$

$$N = (1-u)\int_\mu^\infty k\,\mathrm{d}H(x) = [1 - H(\mu)]\,K_e. \tag{17}$$

7.3 Pareto shocks ⇒ exact Cobb–Douglas

Assume G is Pareto:

$$G(x) = 1 - \left(\frac{\varepsilon}{x}\right)^\alpha, \qquad x \geq \varepsilon, \quad \alpha > 1,$$

and $R \geq \varepsilon$. Then H is also Pareto with scale R (paper Eq. (21)):

$$H(x) = 1 - \left(\frac{R}{x}\right)^\alpha, \qquad x \geq R.$$

One can compute

$$1 - H(\mu) = \left(\frac{R}{\mu}\right)^{\alpha}, \qquad \mathbb{E}[x \mid x \geq \mu] = \frac{\alpha}{\alpha - 1}\mu.$$

Plugging into (16)–(17) gives (paper Eq. (22)):

$$Y = AK_e^{\gamma} N^{1-\gamma}, \qquad \gamma = \frac{1}{\alpha}, \tag{18}$$

with TFP (paper Eq. (23)):

$$A = \frac{R}{1-\gamma}. \tag{19}$$

7.4 Interpretation

- A is increasing in R: tougher selection (higher destruction threshold) raises average productivity among surviving matches.
- This is a model where "TFP" is not a primitive; it is an *equilibrium outcome* of labor-market frictions and institutions.

8 Policy experiments: how labor-market institutions move R and A

8.1 Policy instruments

Four policies (all proportional to k):

- unemployment benefit: τ_b (paid to unemployed),
- hiring subsidy: τ_h (paid to firms upon hiring),
- employment subsidy: τ_e (paid per period of employment),
- firing tax: τ_f (paid by firms upon destruction).

8.2 Value functions with policy

$W(x)$ remains as in (5), but (4), (6), (7) generalize to:

$$rU = \tau_b + \theta q(\theta) \int \max\{W(z) - U, 0\}\, \mathrm{d}G(z), \tag{20}$$

$$rV = -ck + q(\theta) \int \max\{J(z) + \tau_h k - V, 0\}\, \mathrm{d}G(z), \tag{21}$$

$$rJ(x) = \pi(x) + \tau_e k + \lambda \int \max\{J(z) - V + \tau_f k, 0\}\, \mathrm{d}G(z) - (\delta + \lambda)\big(J(x) - V + \tau_f k\big). \tag{22}$$

8.3 Closed-form JD and JC under Pareto

Under Pareto G and $q(\theta) = \theta^{-\eta}$, equilibrium is characterized by (paper Eqs. (24)–(25)):

$$0 = R - \varphi - c + \tau_e + r\tau_f - \left(\tau_b + \frac{\beta}{1-\beta}c\theta\right) + \frac{\lambda \varepsilon^{\alpha} R^{1-\alpha}}{(\alpha-1)(r+\delta+\lambda)}, \tag{23}$$

$$0 = \frac{\varepsilon^{\alpha} R^{1-\alpha}}{(\alpha-1)(r+\delta+\lambda)} + \left(\frac{\varepsilon}{R}\right)^{\alpha}(\tau_h - \tau_f) - \frac{c}{(1-\beta)q(\theta)}. \tag{24}$$

8.4 Existence/uniqueness and unemployment effects (paper Proposition 1)

Proposition 1 (paper Prop. 1 (existence/uniqueness and sign results))**.** *Under a mild parameter restriction, there exists a unique equilibrium* (R, θ) *solving* (23)–(24)*. Moreover:*

- R *and* θ *are increasing in the hiring subsidy* τ_h *and the unemployment benefit* τ_b*;*
- R *is decreasing in the employment subsidy* τ_e *and in the firing tax* τ_f*;*
- θ *is decreasing in the firing tax* τ_f*;*
- *unemployment* u *decreases with* τ_f *and* τ_e *and increases with* τ_h *and* τ_b*.*

8.5 TFP effects (paper Proposition 2)

Because $A \propto R$ (Eq. (19)), signs for TFP come from signs of $dR/d\tau_i$.

Proposition 2 (TFP effects of policies (paper Prop. 2))**.** *Employment subsidies* τ_e *and firing taxes* τ_f reduce *TFP* A*. Hiring subsidies* τ_h *and unemployment benefits* τ_b increase *TFP* A*.*

Why the signs make sense.

- τ_e and τ_f make it cheaper to keep low-x matches alive $\Rightarrow$ lower R (less selection) $\Rightarrow$ lower A.
- τ_b raises the worker outside option $\Rightarrow$ higher $R \Rightarrow$ higher A.
- τ_h raises job creation and θ, and thus strengthens the worker outside option; continuation becomes harder $\Rightarrow$ higher $R \Rightarrow$ higher A.

8.6 Figure 1 (shifts of JD and JC curves)

Figure 1 illustrates these shifts in (θ, R) space:

- Hiring subsidies: primarily shift JC outward, raising θ and (via outside options) raising R.
- Unemployment benefits: shift JD toward higher R.
- Employment subsidies: shift JD toward lower R.
- Firing taxes: shift JD toward lower R and also reduce job creation (JC shifts), lowering both θ and R.

9 TFP measurement: A vs. $\hat{A}$ vs. $\tilde{A}$

9.1 True production function uses employed capital and hours

Under Pareto, the *true* aggregate relationship is

$$Y = AK_e^{\gamma}N^{1-\gamma}, \quad \gamma = \frac{1}{\alpha}, \quad A = \frac{R}{1-\gamma}.$$

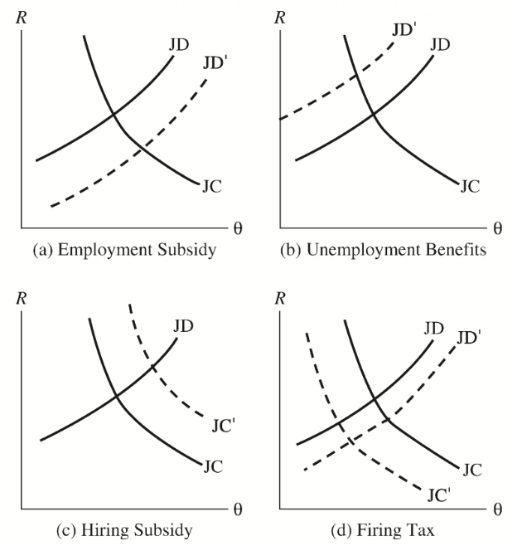

FIGURE 1
Equilibrium effects of policies on job-creation (JC) and job-destruction (JD) rates

9.2 If we use total capital K instead of K_e

A statistician who measures output Y and hours N correctly but uses K rather than K_e would compute

$$\hat{A} \equiv \frac{Y}{K^{\gamma} N^{1-\gamma}} = A \left(\frac{K_e}{K} \right)^{\gamma} = A \left(\frac{1-u}{1-(1-\theta)u} \right)^{\gamma}. \tag{25}$$

So $\hat{A}$ falls when unemployment rises (capital is "stuck" in vacancies).

9.3 If we use employment E instead of hours N

Let $E \equiv 1-u$. From (17) and Pareto algebra, hours satisfy

$$N = \left(\frac{R}{\mu} \right)^{\alpha} K_e = \left(\frac{R}{\mu} \right)^{1/\gamma} \frac{KE}{1-(1-\theta)u}.$$

If the statistician fits a Cobb–Douglas in (K, E) with exponent γ, the implied Solow residual is

$$\tilde{A} \equiv \frac{Y}{K^{\gamma} E^{1-\gamma}} = \left[\left(\frac{R}{\mu} \right)^{1/\gamma} \frac{K}{1-(1-\theta)u} \right]^{1-\gamma} \hat{A}. \tag{26}$$

9.4 Labor hoarding and measured TFP

When $R < \varphi$, there are employed matches with $x \in [R, \varphi]$ that produce zero hours. A convenient hoarding statistic is the mass of employed matches below φ:

$$H(\varphi) = \mathbb{P}(x \leq \varphi \mid x \geq R) = \frac{G(\varphi) - G(R)}{1 - G(R)}.$$

Policies that lower R (e.g. firing taxes) increase $H(\varphi)$: more low-productivity matches survive and hoard labor.

TABLE 1

Baseline results

		$\tau_i/\bar{w}$	D	u	A_{τ_i}/A_0	$\hat{A}_{\tau_i}/\hat{A}_0$	$\tilde{A}_{\tau_i}/\tilde{A}_0$	$H(\phi)$
Baseline		0·11	10	0·069	1·000	1·000	1·000	0·050
τ_b	0·25	0·12	7	0·096	1·023	0·847	0·787	0·017
	0·19	0·10	18	0·038	0·975	1·318	1·497	0·086
τ_f	0·02	0·01	16	0·041	0·977	1·278	1·432	0·082
	0·04	0·02	54	0·012	0·954	1·973	2·717	0·114
τ_e	0·02	0·01	14	0·048	0·983	1·183	1·277	0·074
	0·04	0·02	23	0·030	0·968	1·463	1·745	0·095
τ_h	0·02	0·01	7	0·096	1·023	0·840	0·777	0·017
	0·03	0·02	6	0·108	1·034	0·786	0·706	0·001

10 Quantitative analysis (Table 1)

10.1 Calibration strategy

Key ingredients:

- Matching: $q(\theta) = \theta^{-\eta}$ with $\eta = 0.72$.
- Bargaining: $\beta = \eta$ (Hosios condition), so absent policy the equilibrium is efficient.
- Rates: quarterly $r = c = 0.012$ (so annual discount factor ≈ 0.953).
- Shocks: Pareto with $\alpha = 3/2$, support normalized $\varepsilon = 1$.
- Exogenous separation $\delta = 0$ and redraw intensity $\lambda = 1.23$ are chosen to match job duration/unemployment.
- Initial draw distribution G_0 has lower support $\varepsilon_0 = 0.6$.
- Baseline policy/technology: unemployment benefit $\tau_b = 0.22$ and operating-cost parameter $\varphi = 1.095$.

Baseline equilibrium: $R \approx 1.058$ and average wage $\bar{w} \approx 2.014$.

10.2 Reading Table 1

Table 1 reports unemployment u, job duration D (quarters), "true" TFP A, two measured residuals $\hat{A}$ and $\tilde{A}$, and hoarding $H(\varphi)$. All TFP objects are normalized to 1 in the baseline row.

Baseline. Replacement rate $\tau_b/\bar{w} \approx 0.11$, $D \approx 10$ quarters, $u \approx 6.9\%$, and hoarding about 5%.

Unemployment benefits τ_b. Raising benefits increases R (tighter continuation), which raises true A, but increases unemployment and reduces capital utilization, so $\hat{A}$ and $\tilde{A}$ fall. Lowering benefits reduces R (less selection), lowering true A, but raises utilization so measured residuals increase.

Firing taxes τ_f and employment subsidies τ_e. Both lower R (weaker selection), reducing true A. But they reduce unemployment in the calibration, which increases utilization and raises measured residuals; hoarding $H(\varphi)$ rises sharply.

Hiring subsidies τ_h. They raise R and true A, but in Table 1 they also raise turnover enough that unemployment rises; utilization falls and measured residuals decline.

Key quantitative lesson. Institutions can move true A and measured TFP in *opposite* directions. Hence one can easily mis-rank policies if one uses a naive Solow residual.

11 Extensions and additional aggregation results

11.1 Serially correlated productivity shocks (Markov $G(x'|x)$)

Section 8.1 allows correlated shocks. The selection logic still delivers a stationary distribution H and the same type of JD/JC characterization; the aggregation results continue to rely on the tail behavior of G.

11.2 Endogenous separation rates and Cobb–Douglas (paper Proposition 3)

Section 8.2 lets separations depend on productivity: $\delta(x) = \delta x^{-\sigma}$. The paper shows that for a particular parametric family of G, the aggregate technology is still Cobb–Douglas.

Proposition 3 (paper Prop. 3 (Cobb–Douglas with $\delta(x) = \delta x^{-\sigma}$))**.** *Suppose $\delta(x) = \delta x^{-\sigma}$ and the primitive shocks have density (paper Eq. (36))*

$$\mathrm{d}G(x) = \begin{cases} \dfrac{\alpha\lambda(\alpha+\sigma)\varepsilon^{\alpha}}{\lambda(\alpha+\sigma)+\alpha\delta\varepsilon^{-\sigma}} \left(1+\dfrac{\delta}{\lambda}x^{-\sigma}\right) x^{-(\alpha+1)}\,\mathrm{d}x, & x \geq \varepsilon, \\ 0, & x < \varepsilon, \end{cases}$$

with $\varepsilon, \delta, \sigma > 0$ and $\alpha > 1$. Then in equilibrium, aggregates Y, K_e, N satisfy the Cobb–Douglas form in (18).

11.3 A CES aggregate (paper Proposition 4)

The paper also provides a class of primitive shock distributions that aggregate to a *CES* technology. Specifically, for $\varepsilon > 0$ and $\rho, \sigma \in (0,1)$, let (paper Eq. (37))

$$G(x) = \begin{cases} 0, & x < \varepsilon, \\ 1 - \left[\dfrac{1}{\sigma}\left(\dfrac{x}{\varepsilon}\right)^{\rho/(1-\rho)} - \dfrac{1-\sigma}{\sigma}\right]^{-1/\rho}, & x \geq \varepsilon. \end{cases} \tag{27}$$

Then (paper Proposition 4):

Proposition 4 (paper Prop. 4 (CES aggregation))**.** *If the primitive distribution is given by the above G, then in equilibrium*

$$Y = B\big[\sigma \bar{A} K_e^{\rho} + (1-\sigma)N^{\rho}\big]^{1/\rho}, \qquad B = \frac{1}{1-\sigma},$$

where

$$\bar{A} = \left[\frac{1}{\sigma} \left(\frac{R}{\varepsilon} \right)^{\rho/(1-\rho)} - \frac{1-\sigma}{\sigma} \right].$$

As $\rho \to 0$, this CES aggregate approaches Cobb–Douglas, and the underlying G approaches a Pareto distribution (the paper discusses this limiting connection explicitly).

12 Conclusion

12.1 Summary

Lagos (2006) provides a micro-founded theory in which aggregate total factor productivity (TFP) is **endogenous** and can be **systematically shifted by labor-market institutions**. The central insight is that, in a frictional labor market with search, match-specific productivity risk, and endogenous job destruction, policies affect the **selection margin**—the reservation productivity threshold R below which matches are destroyed. Because R changes the cross-sectional distribution of surviving match productivities, it changes the average productivity of operating production units, and hence aggregate TFP.

Under a Pareto productivity distribution, the model aggregates exactly to a Cobb–Douglas relationship in *employed capital* K_e and *hours* N,

$$Y = AK_e^{\gamma} N^{1-\gamma}, \qquad \gamma = \frac{1}{\alpha}, \qquad A = \frac{R}{1-\gamma},$$

so that **TFP is proportional to the endogenous cutoff** R. Therefore, institutional differences that shift job creation and destruction (and thus R) can generate large cross-economy differences in "technology" as measured by TFP even when primitive technologies are identical.

The paper further shows that **measured** TFP can diverge from **true** TFP because standard Solow residual calculations may use the wrong inputs: total capital K instead of employed capital K_e, and employment E instead of hours N. Since vacancies, idle matches, and labor hoarding respond endogenously to policy, these measurement choices can make the measured residual move *in the opposite direction* from true A.

12.2 Main conclusions

1. **TFP is an equilibrium object.** In this model, TFP is not a primitive; it reflects the equilibrium distribution of match productivities among surviving matches, summarized by the cutoff R.

2. **Labor-market institutions move TFP through selection.** Policies that make it easier to keep low-productivity matches alive reduce selection, lower R, and reduce true TFP A. Policies that strengthen the outside option or intensify competition for labor raise R and increase true TFP.

3. **Unemployment and TFP need not move together.** Some policies raise TFP but also raise unemployment (e.g., by increasing the value of nonemployment), while others reduce unemployment but lower TFP (e.g., by subsidizing match survival).

4. **TFP measurement is not innocuous.** If one computes residuals using total capital K or employment E, the resulting "TFP" can increase even when the model's true productivity term A falls, because utilization/hoarding margins change endogenously.

5. **A unifying explanation for cross-country TFP patterns.** The model provides a coherent channel through which cross-country institutional differences can generate systematic differences in both true and measured TFP, consistent with the idea that long-run income differences are tightly linked to productivity but that productivity itself is shaped by policy.

12.3 Intuition: why the reservation cutoff R is "TFP"

The economics can be summarized by a simple selection logic:

1. Each match draws an idiosyncratic productivity x and is repeatedly hit by shocks that can redraw x.
2. A match survives only if its continuation surplus is nonnegative; this pins down a reservation cutoff R with $S(R) = 0$.
3. Raising R eliminates the bottom tail of matches, improving the average x among those that operate.
4. With a Pareto tail, truncation has a convenient aggregation property: the average productivity among operating matches scales linearly with R, yielding $A \propto R$.

Hence, **TFP is the macro footprint of micro selection**: anything that changes how aggressively the economy destroys low-x matches changes the average efficiency of the surviving production units.

12.4 Intuition: why policies move R in opposite directions

The signs are easiest to interpret by asking whether a policy encourages survival of marginal (low-x) matches or instead strengthens the worker's outside option and intensifies selection:

- **Employment subsidies and firing taxes** make it cheaper to keep a match alive, so the economy tolerates lower-productivity matches $\Rightarrow$ R falls $\Rightarrow$ true A falls.
- **Unemployment benefits** raise the worker's outside option, which raises wage pressure and tightens the continuation condition $\Rightarrow$ R rises $\Rightarrow$ true A rises.
- **Hiring subsidies** raise vacancy creation and market tightness, improving outside options and strengthening selection $\Rightarrow$ R rises $\Rightarrow$ true A rises (even though unemployment may rise depending on general equilibrium).

12.5 Why measured TFP can disagree with true TFP

In the model there are two distinct "hidden" margins that standard growth accounting may mishandle:

1. **Capital employment vs. total capital:** when the vacancy margin changes, the fraction of capital actually deployed in filled jobs changes, so using K instead of K_e contaminates the residual.
2. **Hours vs. employment:** when labor hoarding rises, employment can remain high while hours fall, so using headcount E instead of hours N also contaminates the residual.

Therefore, policy can simultaneously lower true A (via weaker selection) and raise measured residuals (via higher utilization or lower unemployment), creating an apparent empirical puzzle that is resolved once input measurement is aligned with the model's aggregation structure.

12.6 Takeaway

Lagos (2006) shows that aggregate TFP can be an endogenous outcome of search frictions and endogenous job destruction: policies that change the reservation productivity cutoff R change selection and thus true TFP, while standard Solow residuals can be misleading when capital employment and labor hoarding vary endogenously.